FEDERAL SPECIFICATION

CABLE, TELEPHONE (FIGURE 8 DROP)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers 17-1/2 AWG, single pair drop telephone cable and a parallel integral support messenger covered by extruded polyvinyl chloride jacket.

1.2 Classification. Telephone cable furnished under this specification shall be of one type and of one size (see 1.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification:

Federal Specification:

J-C-98—Cable and Wire, Insulated; Methods of Sampling and Testing.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-12000—Cable, Cord, and Wire, Electric; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109—Quality Assurance Terms and Definitions.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Standards Association (ASA) Publications:

C7.1—Soft or Annealed Copper Wire (ASTM B-3).

C8.22—Methods for Testing Rubber and Thermoplastic Wire and Cable (ASTM D-470).

C8.33—Thermoplastic Vinyl Polymer Sheath compound for Electrical Insulated Cords and Cables (ASTM D-1047).

(Applications for copies should be addressed to the American Standards Association, 10 East 40th St., New York, N. Y., 10016.)

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American Society for Testing and Materials (ASTM) Publications:

D 1238—Test for Measuring Flow Rates of Thermoplastics by Extrusion Plastometer.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

National Electrical Manufacturers Association (NEMA) Publications:

WC5-1961—Thermoplastic-insulated Wire and Cable for the transmission and distribution of Electrical Energy.

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th St., New York, N. Y., 10017.)

Rural Electrification Administration (REA) Publications:

PE-8—Zinc-coated (Galvanized) High Tensile Steel Telephone Wire.

PE-200—Polyethylene Raw Material.

(Applications for copies should be addressed to the United States Agriculture Department, Rural Electrification Administration, Washington, D. C., 20250.)

3. REQUIREMENTS

3.1 General. The cable shall consist of a twisted single pair of individually insulated solid copper conductors and a parallel support messenger covered by an extruded black semirigid polyvinyl chloride jacket.

3.2 Material. The materials used for telephone cable shall be as specified herein.

3.3 Conductors. Each conductor shall be solid round wire of commercially pure annealed copper. The conductors shall meet the requirements of ASA C7.1, except that requirements for dimensions and permissible variations are waived.

3.3.1 *Size.* The conductors shall be 17-1/2 AWG having a nominal diameter of 0.0432 inch.

3.3.2 *Joints.* Joints made in conductors during the manufacturing process may be brazed, using a silver alloy solder and a non-acid flux, or they may be welded. In joints made prior to the conductor insulating operation, the ends of the conductors shall be butted. Joints shall be free from lumps and sharp projections. The tensile strength of any section of a conductor, including a factory joint, shall be not less than 85 percent of the tensile strength of an adjacent section of a solid conductor of equal length without a joint.

3.3.3 *Finish.* The conductors shall be free from kinks, scales, inequalities, splints, or other flaws.

3.4 Insulation. Each conductor of the cable shall be insulated with colored, high density polyethylene. The colored polyethylene insulating grade raw material shall meet the requirements of REA PE-200. The conductors in an individual length of cable shall be insulated with the same type of raw material. The insulation shall be smooth and concentric.

3.4.1 *Physical properties.* Specimen of the insulation shall satisfy the following requirements for melt index, tensile strength, elongation, cold bend, and shrink back.

3.4.1.1 *Melt index.* The melt index shall not exceed 0.6 gram (g.) per 10 minutes.

3.4.1.2 *Tensile strength.* The tensile strength shall be a minimum of 2400 p.s.i.

3.4.1.3 *Elongation.* The elongation, at rupture, shall be a minimum of 150 percent.

3.4.1.4 *Cold bend.* The insulation shall show no evidence of cracking.

3.4.1.5 *Shrink back.* When specified, the shrink back, as measured from both ends of the conductor specimens, shall not exceed 1/8 inch (see 6.2 *g*).

3.4.2 *Insulation fault rate.* All individual conductors, prior to jacketing, shall have an insulation resistance of more than 50 megohms. A permissible overall performance level of fault in conductor insulation, based on quality assurance testing, shall average

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not greater than one fault per 20,000 conductor feet.

3.5 Identification of pair.

3.5.1 *Color.* The conductor insulation shall be colored so the "tip" conductor is white and the "ring" conductor is blue.

3.5.2 *Standards of color.* The colors of the insulated conductors shall be in accordance with table I of REA PE-200.

3.6 Cable assembly.

3.6.1 *Twisting of pair.* The insulated conductors shall be twisted into a pair. The twist uniformity shall be designed so as to provide compliance with the capacitance unbalance limit of requirement 3.7.1.

3.6.2 *Core covering.* The use of a core covering over the twisted pair shall be at the option of the manufacturer. If used, it shall consist of a nonhygroscopic material.

3.6.3 *Support wire.* A 0.072 inch, grade 190 galvanized steel support wire shall be provided which shall conform to the applicable requirements of REA PE-8 with the following modifications:

(*a*) The resistivity requirement is waived.
(*b*) The mandrel diameter for the adherence of coating and ductility tests should be 0.160 inch.
(*c*) The wire shall be grade 190 having a minimum breaking strength of 770 pounds.
(*d*) The minimum weight of coating shall be 0.5 ounce per square foot of uncoated wire surface.
(*e*) The permissible variation in the wire diameter shall not exceed plus or minus 0.003 inch.
(*f*) No splices of the support wire shall be permitted after the wire drawing operation. Welds are permitted in rods prior to the wire drawing operation.

3.6.4 *Identification marker.*

3.6.4.1 Each length of cable shall be permanently identified as to manufacturer and year of manufacture by the use of manufacturer's colored identification threads. The identification threads shall be placed under the core jacket.

3.6.4.2 An alternate method of marking may be used when agreed to by the contracting officer.

3.6.5 *Jacket.*

3.6.5.1 The support wire and twisted pair core shall be covered with a homogeneous black semirigid polyvinyl chloride jacket.

3.6.5.2 The semirigid polyvinyl chloride compound used for the jacket shall be formulated to provide good weatherability and low temperature properties and resistance to mechanical damage.

3.6.5.3 The jacket shall provide the cable with a tough, flexible protective coating able to withstand exposure to sunlight and the atmosphere's temperature and stress reasonably expected in installation and service. The jacket shall be free from holes, splits, blisters, or other imperfections. It shall be of uniform thickness as is consistent with good commercial practice.

3.6.6 *Physical properties.* Specimens of the jacketing material shall satisfy the following requirements for tensile strength, elongation, cold bend, impact, dimensions and flame resistance.

3.6.6.1 *Tensile strength.* When specified (see 6.2 *g*), the tensile strength shall be a minimum of 2600 p.s.i. before aging and shall be a minimum of 80 percent of original value after aging.

3.6.6.2 *Elongation.* When specified (see 6.2 *g*), the elongation at rupture shall be a minimum of 200 percent before aging and shall be a minimum of 50 percent of original value after aging.

3.6.6.3 *Cold bend.* When specified (see 6.2 *g*), the jacket of the completed cable, with the support wire removed, shall show no evidence of cracking.

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3.6.6.4 *Impact.* When specified (see 6.2 *g*), the cable shall show no evidence of cracks.

3.6.6.5 *Dimensions.*

3.6.6.5.1 The jacket thickness and other controlling dimensions shall be in accordance with figure 1.

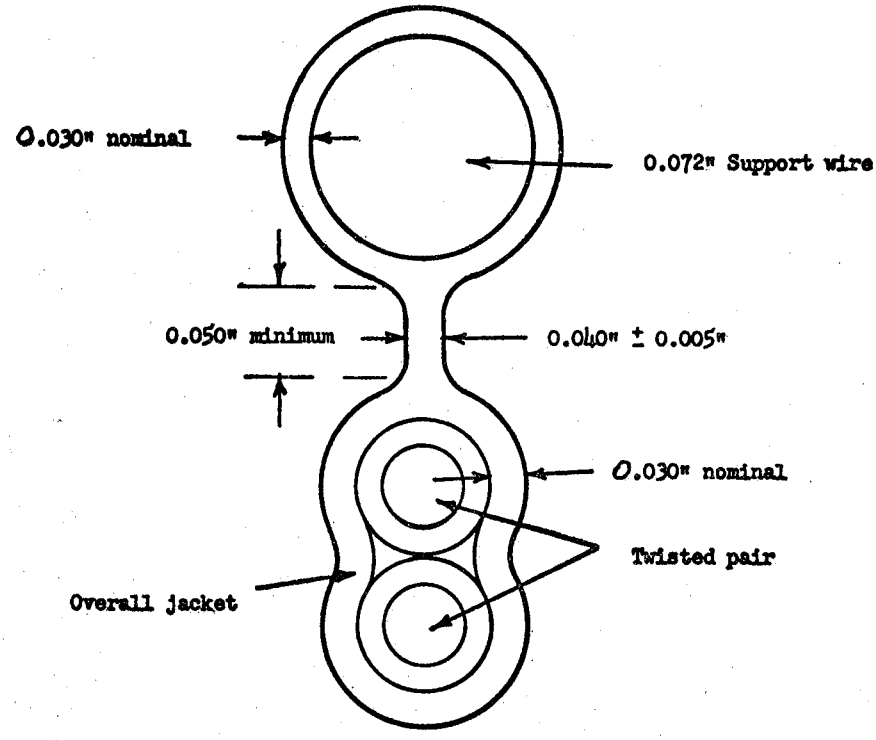


FIGURE 1

3.6.6.5.2 The jacket thickness over the support wire and core shall comply with the following:

Nominal 0.030 inch.
Minimum average 0.027 inch.
Minimum spot 0.021 inch.

3.6.6.5.3 *Adhesion to support wire.* A suitable bond shall be achieved between the jacket and support wire.

3.6.7 *Flame resistance.* The finished cable shall not support combustion for more than one minute.

3.7 **Electrical properties.** Any length of completed cable shall be capable of conforming to the following values for capacitance unbalance, dielectric strength, insulation resistance, conductor resistance, and resistance unbalance.

3.7.1 *Capacitance unbalance to support wire.* The maximum pair to support wire capacitance unbalance for 1,000 foot lengths of completed cable, when measured dry at 1,000 c.p.s., ± 10 percent, shall not exceed 500 m.m.f. per 1,000 feet. The capacitance unbalance is assumed to vary directly with the length of cable.

3.7.2 *Dielectric strength.*

3.7.2.1 In each length of completed cable, the insulation between conductors shall be capable of withstanding for three seconds a d.c. potential whose value is not less than 4500 volts.

3.7.2.2 In each length of completed cable, the insulation between a conductor and the support wire shall be capable of withstanding for three seconds a d.c. potential whose value is not less than 20,000 volts.

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3.8 **Resistance.**

3.8.1 *Insulation resistance.* Each conductor in each completed cable shall have an insulation resistance of not less than 1,000 megohm miles.

3.8.2 *Conductor resistance.* The d.c. resistance of any conductor shall not exceed 7.7 ohms per 1,000 feet measured at or corrected to a temperature of 68°F.

3.8.3 *Resistance unbalance.* The resistance unbalance between conductors shall not exceed 0.7 ohms per 1,000 feet. The resistance unbalance is assumed to vary directly with length.

3.9 **Electrical coordination.** The cable (figure 8 drop) shall not fail under either of the following conditions:

(*a*) Ten minutes having elapsed without starting a flame at the cable (figure 8 drop); or

(*b*) The 19 gauge cable conductors becoming open circuited.

3.10 **Workmanship.** The telephone cable shall be manufactured and finished in a thoroughly workmanlike manner in accordance with all applicable requirements of this specification. Each length of cable shall be free from grounds, short circuits, and open circuits.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Inspection conditions.** Unless otherwise specified herein, all inspections shall be made at room ambient temperature, pressure, and humidity.

4.3 **Definitions.** Definitions of inspection terms used herein shall conform to MIL-STD-109.

4.4 **Acceptance inspection of equipment before preparation for delivery.**

4.4.1 *Group A inspection.* This inspection (including sampling) shall conform to table I, and the ordinary inspection procedures of MIL-STD-105. Level S-4 is to be used.

TABLE I. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical:				
Conductor				
Size	3.3.1	4.5.1.1	1.0	4.0
Joints	3.3.2	4.5.1.1	percent	percent
Finish	3.3.3	4.5.1.1	for	for
Completed cable:			the	the
Dimensions	3.6.6.5	4.5.3.5	group	group
Electrical:				
Capacitance unbalance	3.7.1	4.6	1.0	
Dielectric strength	3.7.2		percent	
Insulation resistance	3.8.1	4.7.1	for	*
Conductor resistance	3.8.2	4.7.2	the	
Resistance unbalance	3.8.3		group	

* All electrical defects are considered major.

4.4.2 *Group B inspection.* This inspection (including sampling) shall conform to table II and to the special procedures for small-sample inspection of MIL-STD-105. Group B inspection normally shall be performed on inpection lots that have been subjected to and met the group A inspection. The acceptable quality levels (AQL's) shall be as shown in table II.

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TABLE II. *Group B inspection*

Inspection	Requirement paragraph	Inspection paragraph	Sampling plan
Insulation:			
Melt index	3.4.1.1	4.5.2.1	2.5
Cold bend	3.4.1.4	4.5.2.4	percent
Jacket:			
Cold bend	3.6.6.3	4.5.3.3	
Completed cable:			Level
Impact	3.6.6.4	4.5.3.4	S-4

4.4.3 *Group C inspection.* This inspection shall be as listed in table III and normally shall be performed on sample units that have been subjected to and have met groups A and B inspections.

TABLE III. *Group C inspection*

Inspection	Requirement paragraph	Inspection paragraph
Tensile strength	3.4.1.2	4.5.2.2
	3.6.6.1	4.5.3.1
Elongation	3.4.1.3	4.5.2.3
	3.6.6.2	4.5.3.2

4.4.3.1 *Sampling for inspection.* Six specimens for each group C inspection shall be selected per 500,000 feet or fraction thereof of finished cable on order, without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

4.4.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure, and shall report to the Government inspector the results thereof and details of the corrective action taken on the process, and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.2).

4.4.4 *Disposition of nonconforming product.* Disposition of nonconforming product shall conform to MIL-STD-105. Such product shall be suitably tagged or identified to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

4.5 Test.

4.5.1 *Conductor.*

4.5.1.1 *Size, joints, and finish.* The size, joints, and finish of the conductors shall be determined in accordance with ASA C7.1 except as specified in 3.3.

4.5.2 *Insulation.*

4.5.2.1 *Melt index.* The melt index shall be determined in accordance with ASTM D 1238 (see 3.4.1.1).

4.5.2.2 *Tensile strength.* The tensile strength shall be determined in accordance with ASA C8.22, except that the speed of jaw separation shall be 2 inches per minute (see 3.4.1.2).

4.5.2.3 *Elongation.* The elongation shall be determined in accordance with ASA C8.22, except that the speed of jaw separation shall be two inches per minute (see 3.4.1.3).

4.5.2.4 *Cold bend.* The cold bend shall be determined in accordance with ASA C8.33, except that the temperature shall be minus 55°C. (—67°F.) ± 1°C. (1.8°F.) and the diameter of the mandrel shall be three times the outside diameter of the insulated conductor (see 3.4.1.4).

4.5.2.5 *Shrink back.* Eight inch specimens shall be cut from the center of a five foot length and then reduced to 6 inches by trimming each end of the specimens. The 6 inch specimens shall be placed in a convection type circulating-air oven at a temperature of 115°C. (239°F.) ± 1°C. (1.8°F.) for a 24 hour period (see 3.4.1.5).

4.5.3 *Jacket.*

4.5.3.1 *Tensile strength.* The tensile strength, before and after aging for 7 days,

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shall be determined in accordance with ASA C8.33, except as specified in 3.6.6.1.

4.5.3.2 *Elongation.* The elongation, before and after aging for 7 days, shall be determined in accordance with ASA C8.33, except as specified in 3.6.6.2.

4.5.3.3 *Cold bend.* The cold bend shall be determined in accordance with ASA C8.33, except that the temperature shall be —40° ± 1°C. and the diameter of the mandrel shall be 8 times the diameter of the minor axis (see 3.6.6.3).

4.5.3.4 *Impact.* The impact resistance shall be determined in accordance with the following: A jacketed cable specimen of approximately 15 inches in length shall be secured across the bottom of a tube 1-1/4 inches in diameter. This assembly is placed on a firm surface in a cold chamber at a temperature of —15°C. (—2°F.) ± 1°C. (1.8°F.) for a period of four hours after which a weight is released at the top of the tube, and the weight that is used shall be such that a 3 pound-foot force is delivered to the jacketed specimen. The weight shall be a cylinder 1 inch in diameter with a flat striking face (see 3.6.6.4).

4.5.3.5 *Dimensions.* The dimensions shall be determined in accordance with figure 1 and J-C-98, method 1018 (see 3.6.6.5).

4.5.3.6 *Adhesion to support wire.* When specified (see 6.2 *g*), the specimen shall be prepared by cutting away the jacketed core from a 9- to 10-inch length of cable leaving only the jacket intact around the support wire. The jacket over the support wire shall be removed for approximately 1 inch at one end, and a reference mark shall be placed on the bare support wire approximately 1/4 inch from the edge of the jacket. The jacket from the other end of the support wire shall be removed so that 6 ± 1/8 inch of jacket are left intact on the support wire. The longer end of the bare support wire shall be passed through a metal die having a minimum aperture diameter of 0.082 inch and a maximum aperture diameter of 0.087 inch. The support wire and die shall be placed in the lower and upper jaws, respectively, of a tensile testing machine. A tension shall be applied between the support wire and die at a speed of jaw separation of 2 inches per minute. The load necessary to break the adhesion shall be recorded as that necessary for the top end of the jacket to first slide over the reference mark. The load required to break the adhesion shall be not less than 50 pounds (see 3.6.6.5.3).

4.5.3.7 *Flame resistance.* The flame resistance shall be determined in accordance with NEMA WC5-1961. The point of application of the test flame should be not less than three inches above the lower end of the sample (see 3.6.7).

4.6 **Capacitance unbalance to support wire.** The capacitance unbalance to support wire shall be determined by the use of a capacitance bridge (see 3.7.1).

4.7 **Resistance.**

4.7.1 *Insulation resistance.* The measurement shall be made with a d.c. potential of not less than 100 nor more than 550 volts applied for one minute. The test may be terminated within the minute as soon as the measurement demonstrates that the specified value has been exceeded (see 3.8.1).

4.7.2 *Conductor resistance.* The conductor resistance shall be determined in accordance with J-C-98, method 6021 (see 3.8.2).

4.8 **Electrical coordination.** The two conductors of a 5-foot length of cable (figure 8 drop) shall be connected in series with two (paired) conductors of a 19 guage, 50-pair plastic insulated conductor cable, by soldered pigtail joints, by suitable splice conductors, or by clamping under the washers of a terminal block. The other ends of the cable (figure 8 drop) conductors shall be terminated at the line binding posts of a fuseless type station protector, with the line binding posts in turn being strapped to the protector ground terminal. The other ends of the cable pair should be connected togeth-

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er. The parallel ends of the cable pair and the protector ground terminal shall be connected to a suitable power source. A 60 cycle current of 120 amperes shall be passed through the series circuit of the cable (paired) and the cable (figure 8 drop). The 5-foot lengths of cable (figure 8 drop) and the 19 gauge cable are to be supported in air free from drafts during the above test (see 3.9).

4.9 Acceptance inspection of preparation for delivery. An inspection shall be made to determine compliance with packaging, packing, and marking requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be level A, B, or C as specified (see 6.2).

5.1.1 *Level A, B, or C.* The cable shall be packaged and packed in accordance with the requirements of MIL-C-12000.

5.2 Marking. The cable shall be marked in accordance with the requirements of MIL-C-12000 with the following additional requirement:

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Civil agency procurement. When lev- B packaging is specified for civil agency procurement, the requirements specified for level A packaging shall apply.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Quantity required.

(*c*) Inspection responsibility, when other than specified (see 4.1).

(*d*) Inspection conditions, if other than specified (see 4.2).

(*e*) Level of packaging and packing required (see 5.1).

(*f*) Special marking, if required (see 5.2).

(*g*) Requirements 3.4.1.5, 3.6.6.1, 3.6.6.2, 3.6.6.3, 3.6.6.4, and 4.5.3.6 are at the option of the contracting officer and must be specified in the invitation for bid if desired. Products meeting this requirement may or may not be readily available at a particular time. Prior to specifying this requirement, the contracting officer shall assure himself that sufficient potential suppliers will respond, without taking exception, to his invitation for bids.

6.4 Guarantee. When desired, provisions for a guarantee or warranty may be included in the invitations for bid. The suggested guarantee is that any length of telephone cable found to be defective in workmanship or material within one year from date of installation, but not more than two years from date of delivery, be replaced.

6.5 It is the intent to use this specification as a replacement for the Rural Electrification Administration specification number PE-46, Figure 8 Drop Wire.

MILITARY INTERESTS:

Custodians:

Army—EL
Navy—YD

User activities:

Army—MI
Navy—MC

Military coordinating activity:

Army—EL

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